

CS490 Lecture Notes

Winter 2025

Instructor: Norhan Abbas

Prepared by: Daisy Ye

Table of Contents

Chapter 1: The Information Systems Strategy Triangle	3
Chapter 2: Strategic Use of Information Resources	8
Chapter 3: Organizational Strategy and Information Systems.....	14
Chapter 5: Information Systems and Digital Transformation	23
Chapter 6: Architecture and Infrastructure	38
Chapter 7: Information Systems Sourcing	44
Chapter 8: Governance of the Information Systems Organization.....	46
Chapter 10: Information Technology Sourcing	55
Chapter 11: Managing IT Projects.....	60
Chapter 12: Business Intelligence, AI, Knowledge Management, & Analytics	67
Chapter 13: IDK the title	72

Chapter 1: The Information Systems Strategy Triangle

IS strategy triangle

- Business strategy
- Organizational strategy
- Information strategy

Strategy = A Game Plan

- a set of coordinated organized set of actions
 - o To fulfill objectives, purposes, and goals.

Sets Limits = Focuses Efforts

- Sets limits on what a business seeks to accomplish

Starts with a mission = The “Why”

- A mission is a clear and compelling statement that unifies a business’ efforts, describes what the business is all about.
- What makes the business unique
- usually abstract

business strategy

- a plan that shows where a business wants to go, and how it expects to get there
- it’s about coordinating actions
- management constructs this plan based on
 - o market force
 - o customer demands
 - o organizational capabilities
 - o vision

organizational strategy

- how the business should be organized, and how it operates to achieve its goals and implement its business strategy
- includes
 - o structures and reporting relationships

- who makes decisions
- control systems (management and measurement)
- performance measurement
- people-based networks
- work distribution
- key business processes
- culture of the place

Information Strategy

- The plan the organization uses to provide information services
- Includes:
 - Technology architecture: the blueprint of how company's technology is organized to set goals.
 - Specific technologies: which tools, software the company uses
 - Platforms: hardware, OS, cloud infrastructure, application platforms
 - Applications: specific software programs the company uses
 - Networks: how computers and systems are connected
 - Data: data being stored and collected

Organizational and information strategy

- The organization strategy:
 - Enables a company to implement its business strategy
 - Helps determine the company's capabilities
- The IS strategy:
 - Enables a company to implement its business strategy
 - Helps determine the company's capabilities

Key points about the strategy triangle

- In successful firms, business strategy drives both organizational and information strategy
 - These firms structure the organization and the information systems to complement the business strategy
- Organizational strategy and IS strategy must complement one another
 - The information systems must support the operations of the business

- Changes in the IS strategy must be accompanied by changes in the organizational strategy
- If a change is made in one corner, it is necessary to evaluate the other corners to maintain alignment

Business models

- It's a framework of how a company creates value, captures value
 - Value Creation
 - Something of value for customers, stakeholders, society, etc.
 - Value Capture
 - Converting the value created into revenue or profits
 - There are different business models
 - A Business Strategy can use multiple models

Business models: primary components (dimensions)

- Value Proposition
 - What product/service, how it helps, who the target customer
- Value Architecture
 - Key resources & activities required for the business to operate & deliver the product/service
- Value Network
 - Partnerships & relationships that the business relies on to succeed
- Value Finance
 - How the business makes money, what the costs to operate

Generic strategies framework

- Strategies for building a competitive advantage
- Three primary strategies
 - Cost leadership
 - Become the lowest-cost producer in the market

- Product/service offered must be comparable in quality to those offered by competitors
- Typically, one cost leader exists, price war otherwise.
- Differentiation
 - Creates a unique product/service
 - A company finds out what customers care about the most, adds value in that area
 - Price charged to customers for the differentiator must seem fair relative to the price charged by competitors.
 - Multiple firms can employ this strategy
- Focus
 - Targets a specific, smaller segment, tailor its offering to that segment
 - Either focus on cost leadership or differentiation
 - Allows the organization to achieve a local competitive advantage even if it doesn't achieve competitive advantage in the market overall.

Dynamic environment strategies

- Porter's Framework focuses on creating competition in stable markets
 - Competitive advantage was sustainable because the rate of change in any industry was relatively slow and manageable
- Newer strategies have since been developed
 - To accommodate the complexity and velocity of the marketplace.
- A hyper competitive market is where
 - Change happens fast
 - Advantages don't last
 - Focus on innovation

Two strategies for dynamic environment

- **The dynamic capabilities model**
 - Dynamic capabilities allow an organization to adapt, thrive, and maintain a competitive advantage in fast-changing environments (such as hyper competitive)
 - How they work? Help organizations:
 1. Sense – detect changes in the business environment, such as shifting market trends, technological advances, or changing customer needs

2. Seize – once detected, seizing involves the capacity to capture and exploit new opportunities that arise from the changes
 3. Transform – reconfigure its internal resources to take advantage of the sensed opportunities
- Resources such as:
 - People (employees, their skills, creativity, experience)
 - Financial (capital)
 - Machines, equipment, and inventory
 - Technology
 - These capabilities are not things that can be bought or acquired; they are part of the organization's experience and capabilities and thus difficult for competitors to imitate
- **The creative destruction model**
- New innovations replace old ones, making older products/services obsolete
 - Drives progress by creating new opportunities while disrupting/eliminating outdated ideas
 - Creation: new products, services
 - Destruction: older products or methods become outdated

Strategic Approach	Key Idea	Application to Information Systems
Porter's generic strategies	Firms achieve competitive advantage through cost leadership, differentiation, or focus.	Understanding which strategy is chosen by a firm is critical to choosing IS to complement the strategy.
Dynamic environment strategies	Speed, agility, and aggressive moves and countermoves by a firm create and sustain competitive advantage.	The speed of change is too fast for manual response, making IS critical to achieving business goals.

Chapter 2: Strategic Use of Information Resources

System hierarchy

- IS are composed of three main elements
 - People - human element
 - Technology - infrastructure
 - Process - structured activities, workflows, procedures
- The middle level represents the integration of people, technology, and processes into information systems
- Management oversees and coordinates the interactions between people, technology, and processes to achieve organizational objectives effectively and efficiently.

Information resources

- A strategic advantage comes from effectively combining financial, production, human, and information resources.
- Information resources include people, technology, processes, and data that help managers perform business tasks.
- Two main types of information resources:
 - IT Assets: Tangible or intangible resources that help create or deliver products/services.
 - Hardware, infrastructure, software, data and information
 - IP, knowledge assets, customer relationships, external
 - IT Capabilities: Skills or knowledge developed over time that enable effective use of IT assets.
- Outsourced IT assets and capabilities contribute to a firm's strategy, even if not owned directly.

Three major categories of IT capabilities

- Technical Skills
 - System analysis, programming, web design

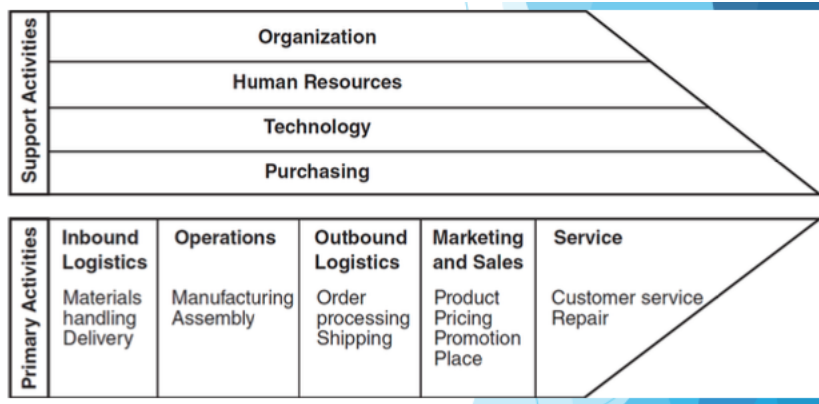
- Developing and integrating applications
- Network design and implementation
- IT Management Skills
 - Providing IT leadership
 - Project management and coordination
 - Understanding business process knowledge
 - Evaluating technology options for business impact
- Relationship Skills
 - External - Managing relationships outside the firm
 - Examples: Market responsiveness, collaborating with platform partners, working with customers/suppliers, negotiating outsourcing contracts
 - Internal - Building trust between IT and business units
 - Examples: Shared IT vision, executive support for IT projects, proactive IT planning

First view - porter's 5 competitive forces

- Threat of new entrants
- Threat of substitute products
- Bargaining power of suppliers
- Bargaining power of buyers
- Industry competitors

Second view - using information resources to alter the value chain

- A second view for describing the strategic use of IS is Porter's value chain
- He divided the activities that create, deliver, and support a company's product or service into two broad categories:
 - Primary activities relate directly to the value created in a product or service
 - Support activities make it possible for the primary activities to exist and remain coordinated
- Each activity may affect how other activities are performed



Using information resources to alter the value chain

- Competition arises from
 - Lowering the cost of activities → Increases profit
 - Adding value to products/services → Justifies higher prices
- Using Information Resources Effectively
 - Reduce costs by optimizing operations
 - Enhance value through better customer insights
- Cost Reduction Alone Isn't Enough
 - Advantage only occurs if costs are lower than competitors'
 - Lower prices attract more buyers and grow market share

The value system

- The value chain model can be extended by linking many value chains into a value system
 - Connected through business relationships and technology
- Much of the advantage of supply chain management comes from understanding how information is used within each value chain and how it can be transferred across value chains within the larger system

Third view - creating and sustaining competitive advantage

- The third view looks at how information resources can be used to attain or sustain competitive advantage
 - The piccoli and ives framework
 - Resources-based view
 - Dynamic capabilities

Piccoli and Ives framework for barriers to competition

- Firms that have created a competitive advantage may try to sustain that advantage as long as possible
 - However, this could be a deadly distraction
 - Competitors might catch up quickly; technologies can be copied or acquired
 - The environment or marketplace might change rapidly
- However, Piccoli and Ives have suggested that a firm that currently has a competitive advantage can attempt to sustain that advantage by raising barriers to erosion of competitive advantage, defined as
 - Impediments to replication of the strategy by competitors
 - In other words, things that the firm can use or do to delay or prevent a competitor from imitating the strategy that led to success

Resource based view

- The resource-based view (RBV) suggests that a firm's competitive position is determined by its unique bundle of resources (assets and capabilities) and how it uses them to create value for the firm
- Competitive advantage arises from resources that are
 - Valuable – enable the firm to become more efficient, effective, or innovative
 - Rare – other firms do not possess it
 - Inimitable – relatively difficult to create or reproduce (imitate)
 - Non-substitutable – unable to replace with something equivalent or similar
 - Non-transferable – difficult to apply to a different firm or environment
- Attributes that help to attain competitive advantage: valuable, rare
- Attributes that help to sustain competitive advantage: inimitable, non-substitutable, non-transferable

Risks

- When information systems and resources are chosen as a tool to outpace a firm's competitors, executives should be aware of the many risks that may surface
- Examples:
 - Awakening a sleeping giant
 - Encountering bad timing
 - Implementing IS poorly
 - Failing to deliver what users want
 - Creating predatory partnerships
 - Running afoul of the law
- With the large expenditure of IT resources needed to create sustainable, strategic advantages, managers should carefully identify and then design mitigation strategies to manage the associated risks

External relationships

- Companies can also participate in a wide range of strategic external relationships
 - A strategic alliance is an interorganizational relationship that affords one or more companies in the relationship a strategic advantage
 - Can take many forms, including joint ventures, joint projects, trade associations, buyer-supplier partnerships, or cartels
- A business ecosystem is a group of strategic alliances in which several partners provide important services to each other and jointly create value for customers
- Co-opetition is a strategy whereby companies cooperate and compete at the same time with companies in their value net
 - The value net includes a company and its competitors and complementors, as well as its customers and suppliers, and the interactions among all of them
 - A complementor is a company whose product or service is used in conjunction with a particular product or service to make a more useful set for the customer

How IT supports external relationships

- IT can be used to support these relationships
 - Digital platforms and marketplaces
 - Connect partners, customers, suppliers

- Facilitate transactions and exchanges
 - Support product development, service delivery
- Data sharing and integration
 - Real-time and secure access to data and information to support decision making
 - Supply chain management
- Facilitate communication and coordination among partners
- E-commerce and digital marketing
 - Online storefronts, payment processing systems, digital marketing platforms

Chapter 3: Organizational Strategy and Information Systems

Organizational Design: 5 Key Questions

1. Who holds decision rights?
2. What are the structures and reporting lines?
3. What networks help get work done?
4. What control systems are in place?
5. What is the culture, values, and beliefs?

Decision Rights

- Authority & responsibility to make specific decisions
- Based on position, knowledge, & scope
- Includes accountability for outcomes
- May involve structured decision-making processes
- Must be flexible & clearly communicated
- Empowering employees boosts ownership & commitment

Decision Rights and IS

- How IS Support Decision Rights
 - Right info to the right person at the right time
 - Data access for evaluation and decision-making
 - Aggregates data from multiple sources
 - Analytics & visualizations = patterns & insights
 - Supports evidence-based management
 - Knowledge management & expert access
 - Decision support tools (simulations, models)
 - Real-time alerts for critical decisions

- More Support Functions
 - Capturing decision metadata (who, when, where, and supporting data)
 - Maintaining audit trails to support accountability and investigations
 - Automating decision-making workflows involving multiple people
 - Enabling collaboration and communication (sharing information, exchanging ideas)
 - Transmitting decisions to those impacted or involved in follow-up actions
 - Automating routine and repetitive decisions using algorithms, rules engines, and AI
 - Ensuring consistency and efficiency in decisions
 - Freeing up time for complex and strategic decision-making

Organizational Structures

- u Organizational structure is a key element of organization design
- u Structures can be hierarchical, flat, matrix, or networked
- u Should support communication and work processes that achieve organizational goals
- u Structure should align with how decision rights are distributed
- u Formal reporting relationships reflect distribution of decision-making authority

Organizational Structures – Hierarchical

Features:

- Division of labour – work segmented into smaller pieces
- Specialization of job functions (e.g., marketing, finance, HR)
- Unity of command – each person has one direct boss
- Small span of control (typically 5–10 direct reports)
- Clear lines of authority and reporting duties
- Centralized decision making, typically at the top

Information Systems (IS) are used to:

- Support departmental functions (finance, HR, operations)

- Store and communicate information to managers at all levels
- Consolidate and manage operational data as it moves upward
- Convey decisions from executives downward
- Organize and manage data centrally

Organizational Structures – Flat

Features:

- Few layers of middle management; broad span of control
- Fluid reporting relationships; less-defined chain of command
- Jobs loosely defined; teamwork and adaptability emphasized
- Direct communication between employees and executives
- Decentralized decision making; employee empowerment
- Focus on performance and results over hierarchy
- Culture that fosters innovation and creativity

Information Systems (IS) are used to:

- Facilitate internal communication and team collaboration
- Provide real-time data and decision support tools
- Support team-based work processes

Organizational Structures – Matrix

Features:

- Employees report to two or more supervisors
- Supervisors manage different aspects of the employee's work
- Shared decision rights between managers
- Flexible sharing of human resources for projects and functions
- Temporary team assignments based on project needs

Information Systems (IS) are used to:

- Simplify operations by sharing information across functions
- Enhance coordination among employees, project managers, and functional managers

Organizational Structures - Networked

Features:

- Interconnected, semi-autonomous “nodes” (teams, units, partners)
- Fluid, dynamic relationships based on needs, expertise, goals
- Flat hierarchies with fewer management layers
- Distributed, decentralized decision-making
- Formal and informal communication networks
- Supports creativity, flexibility, and operational control
- Ideal for dynamic, unstable environments

Information Systems are used to:

- Connect people and processes across the network
- Capture real-time data to support timely decisions
- Enable collaboration across functional boundaries

Emerging Organizational Structures

Types:

- Hybrid structures (vary across departments)
- Adaptive organizations
- Zero-time organizations
- Elastic enterprises
- Heterarchies

Common Themes:

- Flexible and agile
- Rapid reconfiguration of people and resources
- Well-suited for changing environments

Bottom line

- IT can enhance and flatten organizational structures
- Enables open communication and information sharing
- Makes it easier to reorganize and adapt
- Supports business strategy and goal achievement

Informal Networks

- Can be designed or develop naturally
- Help get work done outside formal structure
- Supported by Enterprise Social Networks (ESNs)
- E.g., Slack, Yammer, IBM Connections
- Support communication and collaboration
- Often integrated into work processes

Management control

Management control refers to:

- Establishing processes to monitor, measure, and regulate activities and performance
- Applies to individuals, teams, departments, and systems
- Ensures goals are achieved effectively and efficiently
- Makes sure things happen as planned—on time, in the right way
- Planning – Define how work should be done, how to track it, and what goals to reach
- Monitoring – Measure performance regularly

- Evaluating – Compare what’s actually happening to what was planned
- Compensation/Rewards or Sanctions – Encourage good work, discourage poor performance

Control systems & information systems

- Control systems respond to goals set through planning
- IS help with
 - Data collection – to track progress
 - Evaluation – comparing actual to expected performance
 - Communication – moving info quickly to where it’s needed
- Controls can apply to:
 - Individuals – guiding behavior
 - Automated systems – making adjustments on their own

Technology mediated control (TMC)

- Use of IT to monitor, influence, or regulate people or processes Works in real time with data collection and analysis
- Can apply to people or automated systems
- Focus in this course: control of human behavior

TMC for humans - two approaches

- Managers use tech to monitor and influence employee behavior
- Two main approaches
 - Support–tech gives managers insights to guide workers
 - Automate–tech acts like a manager, guiding workers directly

TMC for humans - algorithmic control

Ø A type of TMC using algorithms to manage human behavior

Ø Algorithms can

Ø Act like a “boss” (e.g., Uber)

- Ø Enforce rules, monitor work, apply consequences
- Ø Or work behind the scenes (e.g., UPS):
 - Ø Monitor performance
 - Ø Give data to managers who take action
- § Algorithms perform management functions:
 - § Allocate resources
 - § Monitor and coordinate work
 - § Optimize tasks
- § Common in HR and gig work platforms
- § Examples: Uber ride matching, Amazon Mechanical Turk

Monitoring work

- IT enables new ways to monitor how work is done
- Ø Detect issues like “cyberslacking”
- Monitoring tools can be built into:
 - Ø Devices, vehicles, cameras, software
- Ø Data collected includes:
 - Ø Keystrokes, time on task, location, break duration
 - Ø Website visits, equipment use
- Monitoring may seem helpful, but it can raise ethical concerns
- Often employees don’t know they’re being tracked
- Raises issues of privacy and fairness
- Managers need to balance control with respect for workers
- When monitoring is appropriate:
 - It should fit naturally into the job
 - Shouldn’t feel intrusive or stressful

- Monitoring that's automatic and job-related is more accepted
- If done poorly, employees may resist or even try to trick the system
- The best monitoring is seamless and reliable

Culture

- Culture impacts IS management and use due to human factors
- A set of shared attitudes, values, and beliefs
- Defines how a group perceives, thinks, and reacts to its environment
- Group can refer to a nation, industry, organization, profession, etc.
- Culture evolves as the group adapts to challenges

Organizational Culture (Schein's Model)

- Edgar Schein's model (1980s): culture has layers
- Artifacts: visible elements (dress, symbols, rituals)
- Espoused values: stated values and behaviors
- Enacted values: values actually seen in practice
- Assumptions: deeply held beliefs, unconscious but powerful

Levels of culture and IT

- Cultures that affect IS
 - National
 - Organizational
 - Subgroup (e.g., IT team, business unit)
- Culture can
 - Influence IS development
 - Affect IT adoption and diffusion
 - Shape IT use and outcomes
 - Impact IT strategy and management

Information systems development

- Cultural clashes between developers and clients are common

- Clients want speed and usability; developers want control and stability
- National culture impacts:
 - o User expectations
 - o Risk perception
 - o Developer-client interactions
 - o UI design and localization
- Organizational culture affects software features and implementation

IT adoption and diffusion

- Risk-tolerant cultures adopt new tech more readily
- Cultures less concerned about hierarchy more likely to share data
- Tech like dashboards or KPIs can challenge hierarchical norms
- In strict hierarchies, broad data access may be resisted or blocked

Chapter 5: Information Systems and Digital Transformation

Manage change in the digital business world

- Change is constant in today's business environment
- To manage well, you must manage change well
- As Information Systems (IS) grow more powerful, the speed and scale of change increases
- IS can either drive success or create problems
- IS can either enable or impede business change
 - The right business process design and the right technology → substantial benefits
 - The wrong business process design or the wrong technology → crisis
- To lead effectively, managers must:
 - Understand how IS support change
 - See the business through a process lens
 - Know how to transform processes the right way

The fundamental perspective of business

- Departments are organized by core functions (e.g., Marketing, Finance, HR)
- Each handles a portion of the overall process
- Benefits:
 - Deep expertise and specialized training
 - Experts can support cross-functional projects
 - Reduces redundancy in roles
- Drawbacks:
 - Information and knowledge often stay siloed
 - Communication gaps between departments
 - Focus on local goals over organizational success

The process perspective of business

- Focuses on end-to-end value creation, not just departments
- Keeps the big picture in view
- A process = a coordinated set of tasks that turns inputs into outputs
- Metrics help measure process performance:
 - Throughput (e.g., 45 units/day)
 - Cycle time (e.g., 28 min)
 - Customer satisfaction (e.g., 4 stars)
 - Profit per output (e.g., \$62)
- Analyzing metrics helps identify ways to improve and optimize

What's a business process

- A business process is a set of coordinated tasks that turn inputs into outputs to achieve a specific goal
- It is interrelated, sequential, and measured with metrics (e.g., time, cost, quality)
- Business processes define how work is done across the organization
- They guide how resources, information, and activities are managed and executed

	Functional Perspective	Business Process Perspective
Definition	Self-contained functional units such as marketing, operations, and finance	Interrelated, sequential set of activities and tasks that turns inputs into outputs
Focus	Function	Cross-function
Goal Accomplishment	Goals optimized for the function, which may be suboptimal for the organization	Goals optimized for the organization, or the "big picture"
Benefits	Core competencies highlighted and developed; functional efficiencies; avoid redundancies by sharing expertise across projects	Avoidance of work duplication and cross-functional communication gaps; organizational effectiveness
Problems	Redundancy of information throughout the organization; cross-functional inefficiencies; communication difficulties	Difficulty in finding staff who can be knowledgeable generalists; need for sophisticated software to coordinate across functional areas

Agile processes

- Built to meet changing customer demands and stay competitive
- Follow a continuous cycle: design → deliver → evaluate → redesign
- The process itself is evaluated and improved — not just the outcome
- Allow for incremental changes and easy reconfiguration
- Common in:
 - Manufacturing– flexible production lines
 - Software development– agile methods & regular retrospectives
- Teams ask: What went well? What didn't? What can we improve?

Dynamic business processes

- Reconfigure automatically using real-time data and adaptive technologies
- Designed to optimize operations in real time
- Common examples:
 - Call Centre Routing
 - Fraud Detection
 - Dynamic Pricing
 - Predictive Maintenance
 - Resource Allocation
- Common traits:
 - Data-driven decisions
 - Self-adjusting through software
 - Focus on continuous optimization

Technology & digitization in agile and dynamic processes

- High IT use enables agility and adaptability
 - More software = easier, faster changes
- Many business processes are becoming 100% digital

- Apps = any kind of software (not just smartphone apps)
- 100% Digital $\neq$ no humans — it means less paperwork, more automation
- Customers and stakeholders expect self-service access
 - Through multiple channels: mobile apps, browsers, desktop apps, IVRs, control panels
- Businesses must adapt or fall behind competitors

Business process redesign

- Two common approaches to redesign:
- **Incremental (adaptive)** – small, continuous improvements
- **Radical (transformational)** – complete rethinking of the process (aka Business Process Reengineering or BPR)
- Both approaches rely on viewing the business as a set of processes, not just departments
- These are tools managers use to drive change and improve performance
- Think of them as two ends of a spectrum:
 - Incremental = gradual gains over time
 - Radical = big improvements, but higher risk and disruption

Incremental changes in business processes

- Improves processes through small, continuous changes
- Common steps:
 - Choose a process to improve
 - Select a performance metric
 - Empower employees to suggest and implement changes
- May involve formal projects or business cases (IT or non-IT)
- Applied to:

- A single process over time
- Or multiple related processes sequentially or in parallel
- Agile development delivers both software and process improvements incrementally

Radical change in business processes

- Incremental change is great for tweaking existing processes
- But it's often not enough for cross-functional or deeply broken processes
- Radical change (aka Business Process Reengineering) enables dramatic, fast improvements
- Aims for breakthrough results on key performance metrics
- Used when:
 - The company is in trouble (e.g., cash flow problems, low morale)
 - There's a shift in business environment (e.g., acquisition, supply chain crisis)
 - The company must outpace competition (e.g., regain market share)

Key aspects of radical change

- Requires major change in a short time
 - Larger impact = more resources available
- Focuses on cross-functional processes
 - Better suited than incremental for big, organization-wide shifts
- Challenges old assumptions
 - Disrupts outdated rules & norms
- Involves networked, cross-functional teams
 - Dynamic roles that can reconfigure easily
- Emphasizes employee empowerment
 - Ownership and accountability support success

- Measures success using business-wide metrics
 - Costs, revenue, customer satisfaction, quality, cycle time, rework rates

Executing Radical Change – Fast and Effectively

- Traditionally takes years and often runs over budget
- But radical change can happen fast when needed
 - COVID online learning transition – weeks, not years
 - Walmart Canada invoicing overhaul – 3 months vs. 18
 - Dutch Railways seat/bike reservation app – built in 4 weeks
- Success factors for fast, radical change:
 - Clear vision
 - Strong leadership
 - Cross-functional stakeholder commitment
 - High technical capability
 - Proactive risk management
 - A real existential threat pushing urgency

Gaming Acceptance of Change

- Change can cause concern or resistance from employees
 - Their daily work is most impacted
 - Resistance may come from fear, confusion, or bad experiences with new Systems
- Incremental change tends to be better received
 - Employees feel ownership and control
 - Improvements often come from grassroots efforts
- Radical change faces more resistance
 - Changes are larger and more disruptive

- Must be carefully planned to gain support and minimize backlash

Managing Change – The Role of Organizational Change Management (OCM)

- OCM = managing the people side of organizational change
- OCM's goals:
 - Consider how change affects employees
 - Help people understand and adapt to the change
 - Reduce stress and uncertainty
 - Ensure smooth and effective implementation
 - Maintain trust with customers, vendors, and partners
 - Reassure external parties about stability
 - Reduce concerns about disruptions to their own operations
 - Preserve loyalty and support, which affects profitability
- Managers can look to Kurt Lewin's and John Kotter's research and publications to
- Understand how people are impacted by change
- Learn how to manage change to achieve positive outcomes
- Figure 5.6 combines Lewin's change model with Kotter's 8 steps

Lewin's Stage	Unfreezing	Changing	Refreezing
Definition	Creating motivation to change	Providing stakeholders with new information, systems, products, or services	Reinforcing change by integrating stakeholders' changed behaviors and attitudes into new operations resulting from change
Kotter's Steps	1. Establish a sense of urgency: Create a compelling reason why change is needed. 2. Create the guiding coalition: Select a team with enough expertise and power to lead the change. 3. Develop a vision and strategy: Use the vision and strategic plan to guide the change process. 4. Communicate the change vision: Devise and implement a communication strategy to consistently convey the vision.	5. Empower broad-based action: Encourage risk-taking and creative problem solving to overcome barriers to change. 6. Generate short-term wins: Celebrate short-term improvements and reward contributions to change effort. 7. Consolidate gains and produce more change: Use credibility from short-term wins to promote more change so that change cascades throughout the organization.	8. Anchor new approaches in the culture: Reinforce change by highlighting areas in which new behaviors and processes are linked to success.

Introduction to enterprise systems

- Enterprise Systems (ES or EIS) = a class of IT tools used to
 - Integrate, coordinate, and share information
 - Across multiple business processes and departments
- Common functional areas integrated by enterprise systems:

- Accounting, HR, Production, Customer & Supplier Management, R&D
- Provide a centralized platform to manage operations across the organization
- Help large companies run hundreds of processes efficiently and in sync

Enterprise System	Sample Processes
Enterprise resource planning (ERP)	Financial management (accounting, financial close, invoice to pay process, receivable management); human capital management (talent management, payrolls, succession planning); operations management (procurement, logistics, requisition invoice payment, parts inventory)
Customer relationship management (CRM)	Marketing (brand management, campaign management); lead management; loyalty program management; sales planning and forecasting; territory and account management; customer service and support (claims, returns, warranties)
Supply chain management (SCM)	Supply chain design; order fulfillment; warehouse management; demand planning, forecasting; sales and operations planning; service parts planning; source-to-pay/procurement process; supplier life cycle management; supply contract management
Product life cycle management (PLM)	Innovation management (strategy and planning, idea capture and management, program/project management); product development and management; product compliance management

Enterprise resource planning (ERP) systems

- ERP = software systems that integrate and manage core internal processes
 - Finance, HR, production, inventory, and manufacturing
- ERP systems have evolved over time, now offering:
 - Access for external stakeholders (partners, customers)
 - Cloud-based deployment (no on-site servers needed)
 - Mobile apps to manage data anytime, anywhere
 - Built-in social and collaboration tools
 - AI/ML features for smarter decision-making and automation
 - Lightweight editions for small to mid-sized businesses

Customer relationship management (CRM)

- CRM = software that helps manage and improve customer relationships
 - Supports sales, support, and service activities
- Helps businesses understand customer needs & behaviors

- Stores data on customers, sales, service interactions, marketing Outcomes
- Improves:
 - Customer service
 - Cross-selling opportunities
 - Sales efficiency
 - Customer retention and revenue
- Goal: Create more effective, personalized interactions using unified customer data
- Leading vendors:
 - Salesforce (24% market share in 2021)
 - SAP, Oracle, and Microsoft (~5% each)

Supply chain management (SCM) systems

- SCM systems integrate processes across:
 - Suppliers, distributors, companies, and customers
- Goal: Optimize costs and opportunities across the entire supply chain
- Enable sharing of real-time data on:
 - Demand, orders, inventory, material availability
- Support supply chain integration through networks and shared protocols
- Help prevent the bullwhip effect (distorted demand signals across the chain)

Supply chain integration challenges

1. Information Integration

- Agree on what data to share, how to format it, and how to secure it

2. Synchronized Planning

- Collaboratively plan forecasting and replenishment decisions

3. Workflow Coordination

- Coordinate and automate processes between partners

- Can range from third-party procurement links to full system integration (e.g., orders and payments)

Advantages of enterprise systems (ES)

- Best practices built in (for processes and data in each industry)
- Modules communicate and share info— one single source of truth
- Centralized operations and real-time decision making across the organization
- Data standardization (clean, consistent, de-duplicated data)
- Standardized procedures across locations for consistency and compliance
- Scalable and flexible— grows and evolves with the business
- Supports innovation and competitive advantage (automation, analytics, insights)

Disadvantages of enterprise systems (ES)

- Data cleanup is intensive
 - Must deduplicate, merge, and clean inconsistent records across departments
- Requires business process redesign
 - Must adapt operations to match system workflows
- Requires organizational changes
 - Roles, responsibilities, and structures must often shift
- High risk of failure
 - Complexity, poor fit, or resistance can derail implementation
- Very high cost
 - Includes software, hidden costs (training, support), and potential business disruption

Enterprise system adoption decision

A key strategic decision when adopting an enterprise system is choosing whether to:

- Let the system drive business process redesign

- Or redesign processes first, then customize the system to match them

When should an enterprise system drive process redesign?

- Good fit when:
 - Organization is just starting out– no established processes
 - Operational processes are not a strategic advantage
 - Current systems are in crisis– not fixable in time or with available Resources
- Poor fit when:
 - Business processes are a source of competitive advantage
 - There is a poor match between the ES and the organization's needs
 - There's no top management support for change
 - Company is growing or changing too rapidly
 - Business requires strategic flexibility
 - The organization is highly decentralized

When enterprise system should NOT drive process redesign?

- Processes are a source of strategic advantage
 - Standard ES may erase differentiation; customization is costly and hard to maintain
- The ES package doesn't fit the business's needs
 - Unique or complex processes may not align with off-the-shelf systems
- The organization isn't ready or suited for ES
 - No executive support
 - Rapid growth makes ES hard to keep up
 - Need for strategic flexibility
 - Decentralized structure with independent business units

Why managers need a business process perspective?

- Avoid relying only on a functional perspective
- Functional silos cause problems:
 - Information stays within departments→ other teams must recreate it
 - Communication gaps→ info can be lost as it moves across functions
 - Departments focus on local goals, not the organization's big-picture goals

What are business processes?

- A business process is a set of interrelated, sequential activities
 - Coordinated to achieve a specific organizational goal
- Represents the workflow of how work is done in an organization
 - Involves resources, information, and tasks
- Goal-oriented by nature
 - Designed to deliver value: serve customers, generate revenue, increase efficiency

Characteristics of business processes

- Cross-functional
 - Span multiple departments and require collaboration among people, systems, and resources
- Structured and repeatable
 - Follow a consistent, documented sequence of tasks to ensure reliability
- End-to-end perspective
 - Cover the full lifecycle—from initiation to completion—including all touchpoints and steps
- Agile and adaptable
 - Evolve in response to external and internal changes (market, tech, regulations, org shifts)
- Measurable and improvable
 - Use KPIs and metrics to track performance and identify opportunities for improvement
- Data-driven

- Depend on data to guide decisions, monitor outcomes, and optimize the process

Techniques for understanding business processes

- Before improving a process, you must understand the current one
- Two key activities:
 1. Documenting processes and workflows
Map out how the process works, step-by-step
 2. Organizing and managing processes
Use tools to structure, analyze, and prepare for change

Documenting processes and workflows

- A workflow = a series of connected tasks performed by people and systems to complete a business process
- Process mapping (or modeling) = visual representation of a workflow
- Common workflow diagramming techniques:
 - Flowcharts
 - Data Flow Diagrams (DFD)
 - Unified Modeling Language (UML)
 - Integrated Definition (IDEF)
 - Business Process Model and Notation (BPMN): Previously called Business Process Modeling Notation

Business process modeling and notation

- BPMN is a widely adopted, comprehensive process modeling standard
- Includes 100+ unique symbols, each with a specific purpose
- BPMN models can be executed by a workflow engine
 - Includes technical attributes for automation
- Looks simple, but takes time and practice to master

Business process development

- Improving a process using IT?
 - Automate steps, reduce duplicate data entry
 - Add IT-based monitoring (e.g., dashboards, KPIs)
- Documenting an existing process?
 - Standardize workflows and improve consistency
 - Train staff or identify inefficiencies
 - Meet regulatory or certification requirements
- Creating a new process?
 - Triggered by new strategy, goals, or business models
 - Often needed for new markets, products, or distribution channels

1- Review the current process

Interview stakeholders to find what works and what doesn't

2- Define process goals

Identify measurable business objectives (e.g., speed, cost, quality)

3- List activities & spot inefficiencies

Analyze tasks—what can be automated, optimized, or eliminated

4- Develop a process map

Create a visual workflow showing the sequence, handoffs, and timing

5- Assign tasks & select managers

Define responsibilities (including for automated tasks)

6- Test and implement

Run small-scale tests to find bugs or gaps before full rollout

7- Measure business value

Compare outcomes to original goals (ROI, efficiency, etc.)

8- Gather feedback & improve

Monitor performance, collect feedback, and refine continuously

Business process management (BPM)

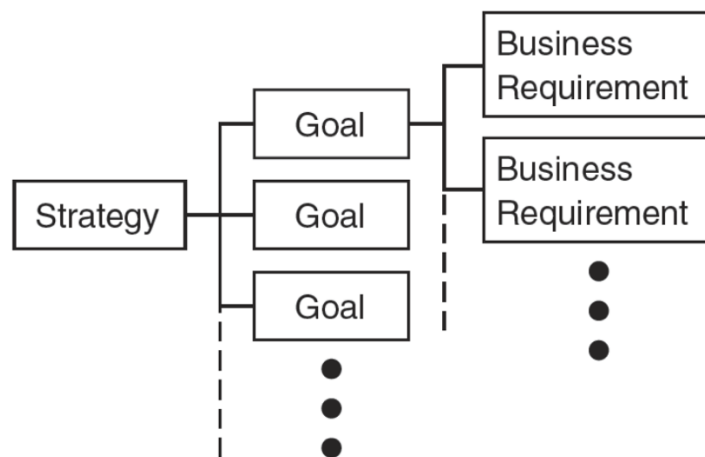
- BPM uses software tools (BPM suites) to model, automate, monitor, and optimize business processes
- Key features of BPM suites:
 - Process mapping & simulation– Visualize and test processes before implementation
 - Automation & execution– Workflow engine runs processes, manages tasks, and triggers actions
 - Monitoring & analytics– Real-time tracking, dashboards, KPIs, heat maps, and insights
 - System integration– Connects with platforms like Salesforce, Oracle, and APIs
 - Code generation & forms– Generate code or UI elements to connect with workflows
 - Human task management– Handles manual steps, queues, and agent assignment
 - Decision management (DMN)– Models rules and logic for automated decisions
 - Versioning & migration– Manage updates and migrate running processes safely

Chapter 6: Architecture and Infrastructure

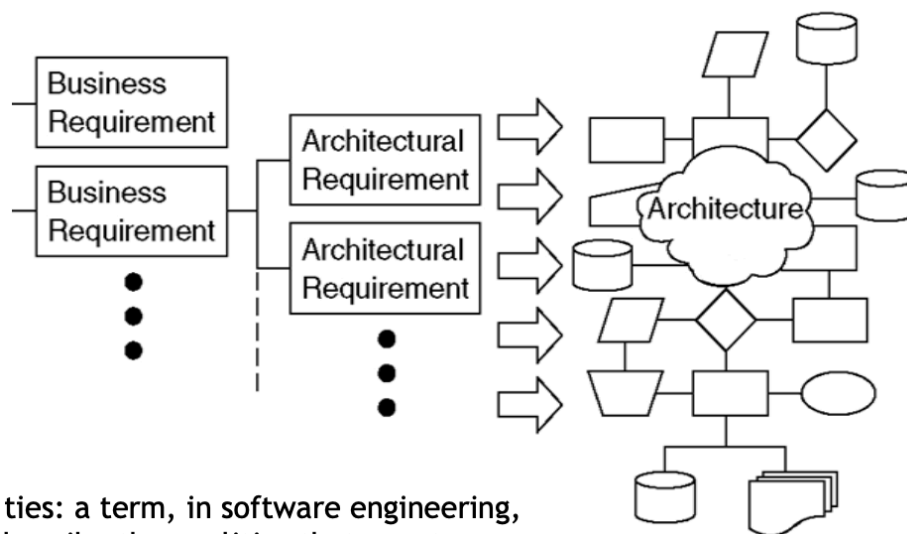
Manager's role in the process

- Reasonable Expectations
- Clearly Communicate the Business Vision
- Adjust Business Plans when needed
- Ensure IT Plans Support Business Objectives

From strategy to business requirements



Business requirements to architecture



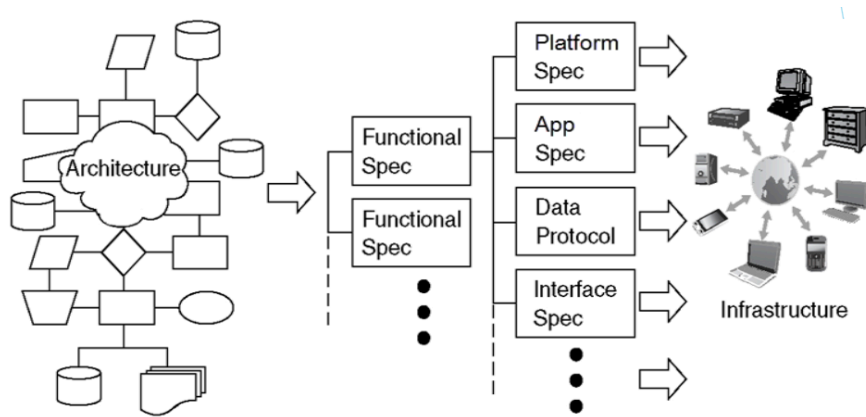
ties: a term, in software engineering, describe the qualities that a system

Architecture levels of abstraction

Conceptual -> logical -> physical

	Conceptual	Logical	Physical
Purpose	High-level, abstract view. Captures fundamental ideas, concepts and relationships. Focuses on <u>what</u> the solution does.	Defines major components, interactions, data flows, processes, system boundaries. Everything needed to complete the solution. Focuses on <u>how</u> the solution works.	Translates logical into concrete implementations. Defines the infrastructure needed to operate.
Abstraction Level	Highly abstract. Easy to understand. No product or technical details.	Focuses on functionality and organization of all system elements. Includes technical details. No product details.	Highly detailed and specific. Actual infrastructure components, configurations and deployments.
Representations	Diagrams, models, or narrative descriptions represent key concepts and relationships. Examples: component diagrams, high-level data or process flow diagrams, conceptual models, textual descriptions, main data entities.	Technical diagrams, models, specifications, and narratives show and describe all components, connections, and locations that are part of the solution. Examples: system diagrams, data flow diagrams, process flow diagrams, details of data entities.	Detailed diagrams, schematics, and specifications provide a detailed view of the actual hardware, software, servers, apps, databases, networks, and other physical components and their locations. Examples: deployment diagrams, network diagrams, server configurations.
Intended Audience	Non-technical stakeholders.	Technical managers, business analysts, system analysts.	Technical managers, IT professionals, engineers.

Architecture to infrastructure



Infrastructure considerations

- Important considerations:
 - Choice of platforms
 - Choice of applications and services
 - Buy vs. build

- Open source vs. proprietary
 - Popular and very capable open-source software: Android, Apache, Linux, Mozilla, NGINX, PERL, PNG, PostgreSQL, Python, and TensorFlow
- Physical location of components and data
- Required connections, firewalls, virtual networks
- Access controls
- Who is responsible for operations and maintenance

Assessing the architecture and infrastructure for future needs

- Once the architecture and infrastructure have been designed, it's important for the manager to assess it
- Why?
 - It's important for the manager to assess it from the perspective of possible future needs.
 - Things might have changed by the time the architecture and infrastructure are implemented.
 - Assess it from the tech, financial, and strategic timeframe perspective

Assess existing architecture

- The infrastructure, architecture currently in use vs the goals and needs of the business today
- Assess the existing systems with the current business strategy.
- Structured Approach to Assessment
 1. Analyze existing architecture & infrastructure
 2. Analyze strategy supported by existing systems
 3. Analyze how the existing architecture and infrastructure fit the current strategic goals.

Assessing strategic timeframe

- Questions to ask
 - how far into the future the business strategy extends?

- How long can the existing arch/infra support strategic goals?

- Strategic Timeframe

1. Varies from industry to industry

- Level of commitment to fixed resources
- Maturity of the industry
- Cyclicalities
- Barriers to entry

2. Varies from company to company

- Reliance on IT
- Rate of IT advances

Assessing adaptability

- Ability to handle expected tech advances
- Ability to absorb unexpected tech leap
- Key Guidelines
 - Loosely coupled systems
 - Set clear boundaries
 - Avoid bleeding-edge technology

Assessing scalability

- The ability of an infrastructure to adapt to increased or decreased demand.
- Why it Matters?
- Key Questions
 - What is the projected growth?
 - Could there be sudden peaks in demand?
 - What if growth never materializes?
 - How does the architecture support changing loads?
- Scalability Techniques
 - Vertical Scalability (Scaling Up)
 - Horizontal scalability (Scaling Out)

Assessing maintainability

- The ease and speed of restoring a system to fully operate after a failure
- Why it Matters?
- Key Considerations
 1. Complexity
 2. Availability of Service people and support
 3. Turnaround time for issue resolutions
 4. Obsolescence Risk

Assessing security & financial aspects

- Security
 - o Security for information systems will be covered in chapter 7 (next week)
- Financial
 - o Valuing IT investments will be covered in chapter 8 in the following week

Enterprise architecture (EA)

- A conceptual blueprint defines how the entire company operates and interacts with its IT systems.
- Aligns IT with business goals to ensure smooth operations and meaningful outcomes.
- Includes the entirety of the business, not just IT, to ensure the entire company is aligned with digital transformation.
- EA is more how the company operates than how the tech is designed.
- EA should be more than automating or improving efficiency
- **Scope of EA**
 - Single organization
 - Collection of organizations
 - Including partners, suppliers, customers

Why create an EA?

- Support digital transformation
- Help departments understand the big picture.
- Identify challenges and risks
- Unify and coordinate processes
- Identify gaps and opportunities –review business capabilities

EA framework

- TOGAF (The Open Group Architecture Framework)
- Zachman Framework for Enterprise Architecture
- FEAf (Federal Enterprise Architecture Framework)
- Gartner
- UAF (Unified Architecture Framework)
- Agile Architecture Framework

EA layers

- Business layer • Applications • Information • Technology

EA challenges

- It is a sophisticated approach usually involving outside consulting agencies to help
- Often leads to a requirement to change the organization design, which companies may be hesitant to undertake
- Produces lots of documentation
 - Difficult to maintain and keep up to date
- Long process: may be out of date before it's completed
- High probability (40% - 90%) of not realizing the major benefits if not carefully planned and implemented

Chapter 7: Information Systems Sourcing

NIST cybersecurity framework (CSF)

- Developed by the U.S. National Institute of Standards and Technology (NIST)
- de facto standard
- Why important?
- Core Functions



Cybersecurity agency in Canada

- Canadian Center for Cyber Security (CCCS)
 - o A unified source of expert advice, guidance, services, and support for:
 - Individuals
 - Small and medium businesses
 - Large organizations and critical infrastructure
 - Government institutions and academia
- Responsibilities? Monitor threats, protect critical infrastructure, coordinate national response

How do you know how secure your organization is?

A balanced Scorecard

1. Technology Risk Measures

2. People/Organizational Risk Measures

3. Supply Chain Risk Measures

4. Financial Risk Measures

Chapter 8: Governance of the Information Systems Organization

IT is more than just technology

- IT not just about technology, it must align with the business goals.
- IT leaders must align technology with business goals.

Success in IT is not just about completing projects. It's about delivering business value.

IT maturity model

1. Level 1 - cost center / basic IT / order taker
 - a. Focus: fixing problems, reducing costs
2. Level 2 - a tool to improve business processes / process improvement / solution provider
 - a. Focus: making business operations more efficient
3. Level 3 - a business partner
 - a. Focus: driving business growth and innovation

Companies that treat IT as a business partner (Level 3) gain a competitive edge.

Maturity Level	Nature of the Level	Engagement Characteristics
Level 1: Business efficiency	IT as order taker	• Reactive • Inside-out • Technology centric • Framed in a context of cost
Level 2: Business effectiveness	IT as solutions provider	• Active • Process centric • Focused on solutions • Framed in a context of projects
Level 3: Business transformation	IT as business partner	• Proactive • Outside-in • Relationship centric • Focused on business growth • Framed on a context of business value and value co-creation

Outsourcing IT organization activities

- What is Outsourcing?
- Why do companies outsource?
- Typically outsourced

- Developing and maintaining systems
- Operating the data center
- Managing internet and network systems

What IT does NOT do

- IT does NOT perform core business functions
 - Selling
 - Manufacturing
 - Accounting
- IT should NOT make key business decisions alone
 - Which projects receive IT funding
 - The level of IT services and security needs
 - Critical technology investments

IT should provide recommendations, but business leaders should make final decisions on the previous

Who should make the decisions?

- IT decisions shouldn't be made by one person –not even the CEO
 - C-level executives should not decide on IT alone
 - IT decisions need input from both IT and business units
 - A formal structure that specifies how IT decisions are made, carried out, reinforced, and challenged should be established

Chief information officer (CIO)

- Business technology strategist
- What does the CIO do?
 - Leads IT strategy
 - Manages IT investments
 - Encourages innovation
 - Works with executives
 - May delegate actual day-to-day management to other IT managers
 - Perform both strategic tasks and operational tasks during a typical day.

- Operational activities include things like identifying and managing the introduction of new technologies into the firm, negotiating partnership relationships with key suppliers and platform partners, and managing the overall IT budget.

CIO's focus

- Reporting
 - Early days: reported to the CFO
 - Now: reports to the CEO
- CIO's focus has shifted
 - From efficiency to effectiveness in a constantly changing marketplace
 - Towards helping executive team formulate business strategy

Importance of business case

- IT projects exist to meet business needs
- Budget constraints require careful selection
- Managers must prove IT is a good investment
- The business case provides structure
 - A formal document that presents all relevant details of a project.
 - Helps executives make a go/no-go decision based on facts, not assumptions.

Realities of IT investments

- A study by Ward, Daniel, and Peppard found that 96% of companies require a business case for IT investments, yet only 30% of these investments succeed in delivering the expected benefits.
- Many IT projects fail because the benefits were never achievable, were exaggerated to secure funding, or there was insufficient understanding of the business changes required for success.
- Companies often overestimate benefits and underestimate costs, leading to unrealistic expectations.
- Management frequently does not demand strong evidence to support major IT investments, resulting in poor decision-making and wasted resources.

IT portfolio management

- Evaluating and managing IT investments collectively to maximize business value.
- IT investments are viewed as a cohesive set of assets.
- The goal
 - o Fund and invest in the most valuable initiatives which, taken together as a whole, bring maximum benefits.

Types of IT investments

- IT projects fall into four main types
 - o **Transactional Systems** – help reduce costs and make processes more efficient
 - o **Infrastructure Systems** – provide basic IT services like internet and networks
 - o **Informational Systems** – help businesses make smart decisions using data
 - o **Strategic Systems** – help businesses gain competitive advantage
- How businesses use IT investments
 - o Companies allocate IT investments differently based on strategy.
 - o Cost-focused firms invest more in transactional systems.
 - o Agility-focused firms prioritize infrastructure investments for flexibility.

Valuing IT investments

- Determining the true value of IT investments is challenging:
 - o IT costs are complex
 - o Infrastructure investments have longer or unclear payback periods
 - o Some IT upgrades are mandatory
 - o Many IT benefits are intangible
 - o IT's impact is often observable, not always measurable

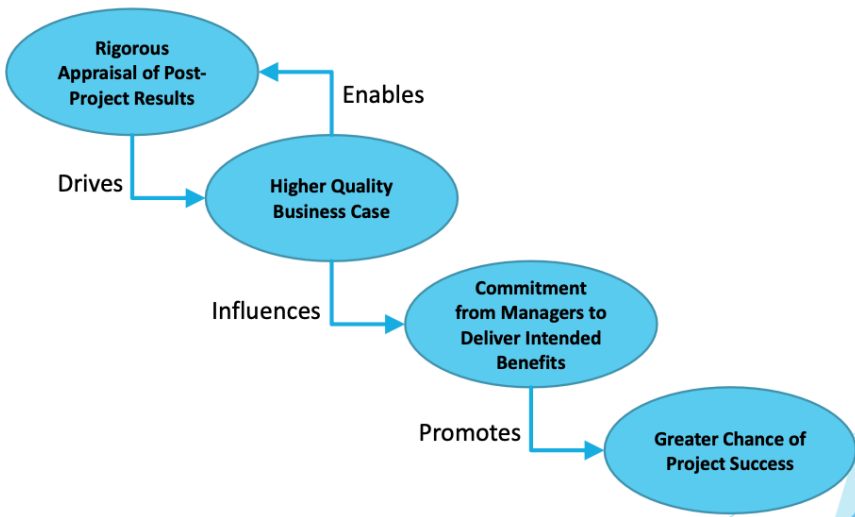
$ROI = (\text{revenue} - \text{investment}) / \text{investment}$

Monitoring IT investments

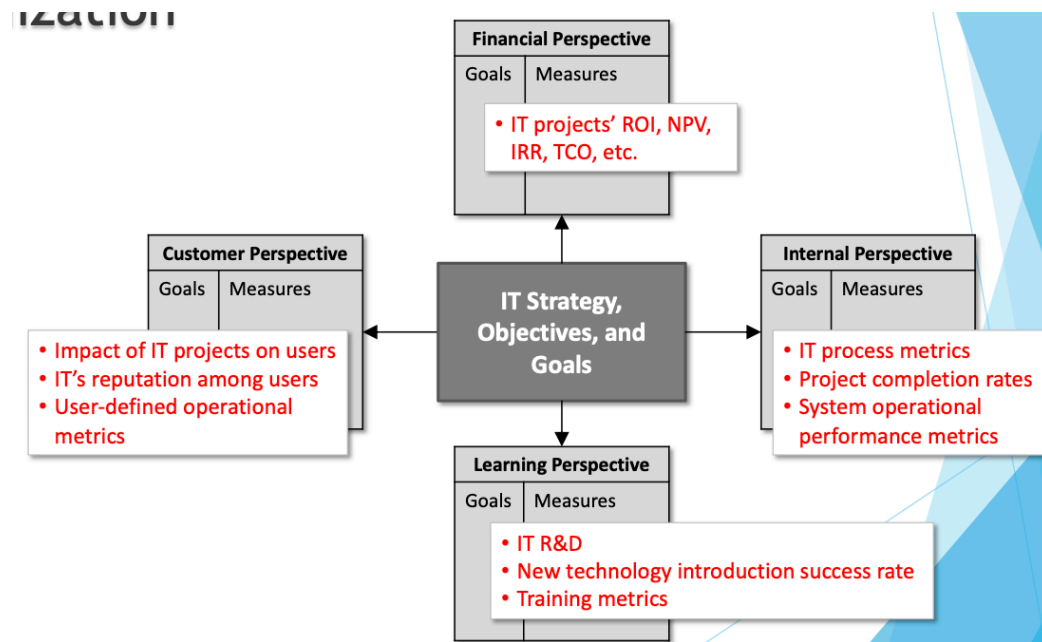
- IT investments must deliver value to the business

- "If you can't measure it, you can't manage it"
- Step 1: Define a common set of metrics
- Step 2: Make metrics visible and communicate them
- Reviewing IT investments after project completion ensures that expected benefits were actually achieved
 - Post-Project Benefit Assessment
 - Many Companies Skip This Step
 - Successful Firms Prioritize Post-Implementation Reviews
- A Rigorous Business Case Approach Leads to:
 - Better business cases
 - Stronger commitment
 - Higher project success rates

Business cases and project success



The balanced scorecard for the IT organization

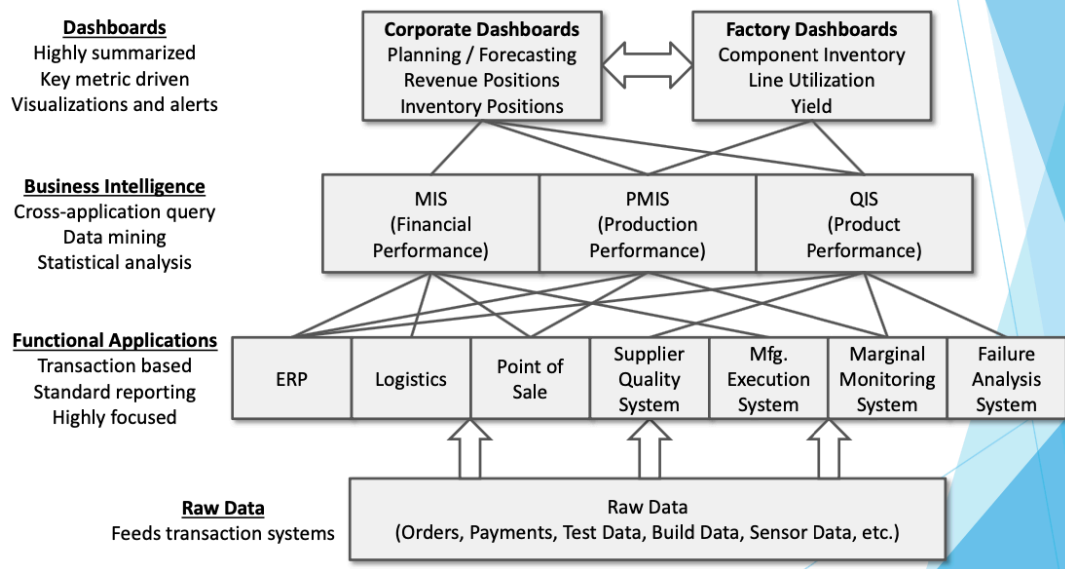


IT dashboards

- Snapshot of metrics at a given point in time (often “right now”)
- Offers “at a glance” picture of how things are going
- Often colors depict conditions
 - o Green: performance within acceptable range
 - o Yellow: performance is slightly outside acceptable range
 - o Red: significant deviance from acceptable range
- Scorecards vs. dashboards:

Scorecards	Dashboards
Long-term strategic goals	Short-term operational goals
Periodic updates, indicating progress towards a target	Real-time performance metrics

Architecture for dashboards



Comparison of IT funding methods

Funding Method	Description	Why do it?	Why not do it?
Chargeback	Charges are calculated based on actual usage.	Fairest method for recovering costs since it is based on actual usage.	IT department must collect details on usage, which can be expensive and difficult. IT department must be able to defend charges, which takes time and resources.
Allocation	Expected IT expenditures are divided by non-usage basis (e.g., number of login IDs, headcount, number of workstations, etc.).	Less bookkeeping for IT. Rate is set once per fiscal year. Basis is well understood. Monthly costs for business units are predictable.	IT department must defend allocation rates. Users can question rates & basis of allocation. Charges may not be proportional to usage rates.
Corporate Budget	Corporate allocates funds to IT in annual budget - to general P&L	No billing to the business units. No rates to compute. IT exercises more control over what projects are done. Encourages use of new technologies.	Have to compete with all other budgeted items for funds. Potential for over-usage by business units. IT group may be less end-user or customer oriented since business units are not footing the bill.

How to determine IT cost

- Basic method

- Add up costs of all the components involved: hardware, software, network, vendor fees, people, etc.
- Other methods
 - Activity-based costing (ABC)
 - Total cost of ownership (TCO)

Total cost of ownership (TCO)

- TCO is a technique used to analyze all the costs of an investment over its lifetime
 - Initial capital expenditures
 - Ongoing operational costs
- Includes costs often forgotten, such as
 - Installation; technical support; administration; training; maintenance; downtime; space and energy; system retirement

TCO component breakdown

- Hardware and software purchases usually make up only 20% of the total cost.
- TCO gives a complete financial picture
- Better decision-making – TCO helps companies compare different IT investments by looking at long-term costs and benefits, not just the initial price.
- Avoid financial surprises
- Track and improve over time

Challenges and considerations with TCO

- TCO isn't everything; IT decisions should also consider non- financial benefits – total benefits ownership (TBO):
 - Increased productivity, Increased interoperability, Improved customer service, Higher capacity and scalability, Enabling future opportunities
- No standard methodology
- Hidden operating costs
- Unpredictable rising costs
- Upgrade costs
- Complexity

Reducing IT costs

- Centralization and standardization of hardware and software u Increase efficiency of supply chain
- Consolidate smaller contracts
- Compare to industry standards
- Focus on higher-than-average IT spending areas
- Virtualization
- Cloud computing
 - o From on-premise infrastructure to on-demand cloud services
 - o Shift from capital expenditures to operating expenses u
 - o Cost controls are needed to avoid surprises

Chapter 10: Information Technology Sourcing

IT services

- IT organizations provide essential information and technology services.
- How IT Services are provided
 - o Insourcing
 - o Outsourcing
 - o Combination Approach
- Effective Outsourcing is a business partnership

Some lingo to consider

- Insourcing vs Outsourcing
- Onshoring vs Offshoring
- Other sourcing models
 - Rural Sourcing
 - Crowdsourcing
 - Crowdfunding
 - Freelancing
 - Global In-house Center (GIC)

Sourcing decision cycle framework

- How IT services are sourced?
 - o How to source? Insource, Outsource
 - o Where to source? Onshore, Offshore, Cloud, Crowd
 - o If offshoring, where? Nearshore, Farshore
- The outsourcing relationship
 - o Once outsourced, the company -> a client company of the provider
 - o Regular evaluations
 - o (unhappy?) Backsourcing brings services back in-house

Why do companies choose to outsource?

- Cost Savings
 - Lower labour costs, economies of scale, infrastructure saving
- Scalability and Flexibility
 - Variable capacity, increased flexibility
- Increased Efficiency and Expertise
 - Specialized providers, access to special skills, exposure to innovation
- Faster Time to Market
- Focus on Core Business & Strategy
- Potential Financial Benefits

Insourcing drivers and risks

Examples of Reasons to Insource	Associated Risks
Related to company's core competencies	<i>If outsourced:</i> Loss of control over strategic initiatives; loss of strategic focus or competitive advantage
Confidential or sensitive IS services or software development	<i>If outsourced:</i> Competitive secrets leaked; competitive advantage lost
Sufficient time available to complete software development projects in-house	<i>If insourced:</i> Project not completed on time
Expertise to provide service or develop the software is available in-house	<i>If outsourced:</i> Technological innovations limited to what provider offers; overreliance on provider's skills
Unable to find a provider that is reliable and able to provide the services for the duration of the contract	<i>If outsourced:</i> Project not completed, or if completed, project over budget and late when another provider takes over
Unable to find a provider with culture and practices compatible with the company	<i>If outsourced:</i> Conflict between provider and client personnel
Provider can't take advantage of economies of scale to offer services or develop software cheaper than in-house	<i>If outsourced:</i> Excessive costs of project or operations because of the way the contract is written

Outsourcing drivers and risks

Examples of Reasons to Outsource	Associated Risks
Provider has economies of scale that make it cheaper to provide the service or develop the software in-house	<i>If insourced:</i> Higher costs than comparable outsourcing
Insufficient skills and expertise to provide service or develop the software in-house	<i>If insourced:</i> Company is unable to deliver the service or develop the software; loss of business
Provider has better ability to handle peak loads	<i>If insourced:</i> Slowdowns or service interruptions; loss of business
Commodity service or product	<i>If insourced:</i> Waste time, resources, and opportunities developing service or product that could be purchased immediately and cheaper
Insufficient time available to complete software development projects in-house	<i>If insourced:</i> Project not completed on time
Able to find a provider that is reliable , able to provide the services for the duration of the contract, and with culture and practices compatible with the company	<i>If insourced:</i> Required capabilities must be developed in-house before service can be provided; service delivered at higher cost
Opportunity to consolidate data centers	<i>If insourced:</i> Inefficient operations

Outsourcing risks

- Loss of control
- Lack of Technological Innovation
- Loss of Competitive Advantage
- Dependence on the outsourcing provider
- Failure to achieve cost savings

Outsourcing using cloud computing

- Benefits
- No need to own or maintain data centers.
- Pay-as-you-go pricing.
- Easily increase or decrease computing power and storage.
- Best-in-class security controls.
- High availability & disaster recovery for business continuity.

- Services are accessible from anywhere with fast, reliable internet.
- Cloud offers a variety of pre-built services (IaaS, PaaS, SaaS) for different business needs.

Cloud computing risks

- Vendor lock-in
- Network latency
- Security responsibilities
- Migration challenges
- Uncontrolled usage can increase costs
- Rising costs of cloud computing

Selecting an offshore destination

- Language
- Cultural Compatibility
- Political & Security Risks
- Legal & Regulator Environment
- Infrastructure & Technology
- Process Maturity & Work Quality

Hidden costs of outsourcing

- Preparation costs
 - Research & Analysis
 - Request for Proposals
 - Vendor Selection
- Transition costs

- Knowledge Transfer
- Business Process Changes
- Possible Layoffs & HR Costs

Chapter 11: Managing IT Projects

Why many IT projects fail

- Failure to deliver expected benefits
- Weak project management & leadership
- Poor stakeholder engagement
- Insufficient or mismatched resources
- Remote work & collaboration challenges

Projects vs. Operations

Project

Common Planned Executed Controlled Performed by people Requires resources	Temporary Duration Ongoing	Reach a goal or accomplish a task Purpose Sustain the enterprise	Unique Product or Service Repetitive	Informal Quality Control Formal	Goal reached or task accomplished Trigger to Change Operations no longer serve goals
---	--	--	--	---	--

Operations

Projects stakeholders

- Anyone involved in the project or affected by its outcome.
- Project Manager – oversees and ensures the project stays on track
- Project Team – people actively working on the project
- Project Sponsor – general manager, funds the project, expects to use its results
- Customers
- Employees

Project management

- The application of knowledge, skills, tools, and techniques to project activities to meet requirements.



- Requires managing trade-offs

Project management office (PMO)

- A department that
 - Boosts efficiency
 - Gathers expertise
 - Improves project delivery
 - Responsibilities
- Project Management Expertise
 - Processes & Methods
 - Training & Mentoring
 - Project Management Tools
 - Portfolio Management

Essential elements of a project

- Project Management
 - Project Sponsor
 - Project Manager
- Project Team
- Project Plan
- Common Vocabulary

Project sponsor

- Initiates, funds, supports the project at a high level
- Must be available throughout the project
- Main responsibilities
 - Ensures there's enough money, staff, tools for the project
 - Provides leadership and resources (people & financial)

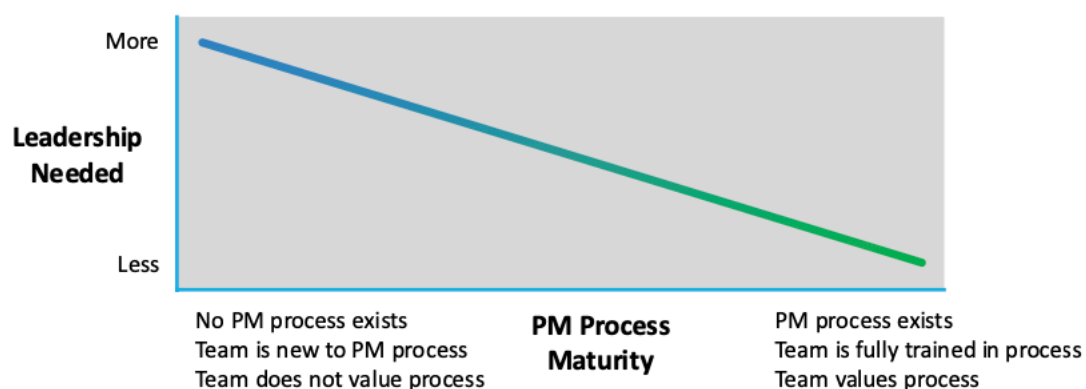
- Escalates problems
- Signs off on scope, budget, and major changes.
- Builds visibility of the project
- Liaison between the project team and other stakeholders

Project manager

- Responsible for planning, executing, and controlling the project
- Main responsibilities
 - Clearly outline what needs to be done
 - Ensures tasks are completed on schedule
 - Organizes and coordinates team members
 - Adjusts on scope, cost, and time to maintain quality.
 - Takes corrective actions to keep project on track
 - Measures, tracks, report progress

Project leadership

- Lack of leadership leads to unmotivated or confused people doing the wrong things and ultimately derailing the project
- Strong project leaders focus, align, and motivate team members
- Strong processes vs. strong leadership:



Project team

- People who work together to complete the project.
- Key Aspects of a strong project team
 - Clear objective
 - Defined role and responsibilities
 - Right skills and capabilities
 - Information sharing
 - collaboration

Project plan

- Outlines tasks, timelines, resources to complete the project successfully.
- Key components
 - Project Schedule
 - Identifies critical start/end dates
 - Breaks work into phases
 - Milestones & Control Gates
- Project schedule can be visualized

Subprojects

- Subprojects may be created to organize the project work around distinct activities or resources
- Examples
 - quality control testing
 - data conversion
 - implementation planning
 - outsourced work
- Subprojects could be contracted out to limit costs or other drains on crucial project resources

Common project vocabulary

- Project teams include members from different backgrounds and parts of the organization
- Technical vocabulary has different meanings to different people
 - o “end of year”
 - o “customer” vs “user” vs “client”
 - o “sale” vs “divestment”
- Create a common project vocabulary
- Record and explain terms used in the project
- Helps avoid misunderstanding among team members and other stakeholders

Programs

- A Program is a collection of multiple related projects managed together to achieve larger business benefits.
- Projects– Focus on delivering a specific outcome (e.g., a new product, service, or system).
- Programs– Focus on achieving business-wide benefits by coordinating multiple projects.

What makes an IT project risky?

- Complexity - More moving parts and interdependencies increase risk
- Clarity - If project goals and requirements are unclear, uncertainty rises
- Size - Large projects with more people, resources, and teams are harder to manage.

Risk factor: complexity

Factors That Increase Complexity

- New or Unproven Technology
- Fast-Changing Business Environment
- Many Interdependent Parts
- Specialized Expertise Required

To Manage

- Experienced Project Team
- Rely on consultants and Vendors
- Improve Internal communication
- Divide work into subprojects

Risk factor: clarity

Factors That Lower Clarity

- Unclear or Changing Requirements
- Conflicting Stakeholder expectations
- Regulatory Uncertainty

To Manage

- Develop a scope management plan
- Develop a change control process
- Identify all stakeholders, understand and balance their goals
- Remind stakeholders of benefits to be gained
- Involve users in the project to help clarify requirements

Strategies to deal with low clarity

- Techniques and processes
 - SDLC:
 - JAR (Joint Application Requirements) / JRP (Joint Requirements Planning) / JAD (Joint Application Design) sessions
 - Focused meetings with all relevant stakeholders
 - Agile:
 - Product backlog items (PBIs) - user stories, features, epics, spikes, etc.
 - Product backlog – all remaining work
 - Backlog refinement – elaboration of PBIs to prepare for development
 - Prioritization – sequencing the PBIs for development (t-shirt sizing, story mapping)
 - Effective product owner (engineering background, field experience, and business experience)

Risk factor: size

Big projects are riskier than small ones

- To Manage
 - Prioritize Requirements
 - Eliminate Unnecessary Functionality
 - Break the Project into Subprojects
 - Use Independent Subsystems (e.g., Microservices)
 - Sustain Commitment Over Time

IT project estimates

- Difficult to estimate time and costs
- Most fail to meet schedule and budget
- Most late projects cannot be sped up by adding more manpower
 - Brooks' law: "Adding manpower to a late software project makes it later"
 - Ramp-up time for new workers
 - More people → more communication to coordinate work
 - But planning for extra manpower up front can work
 - And sometimes you can add more people to a late project and get the work done faster

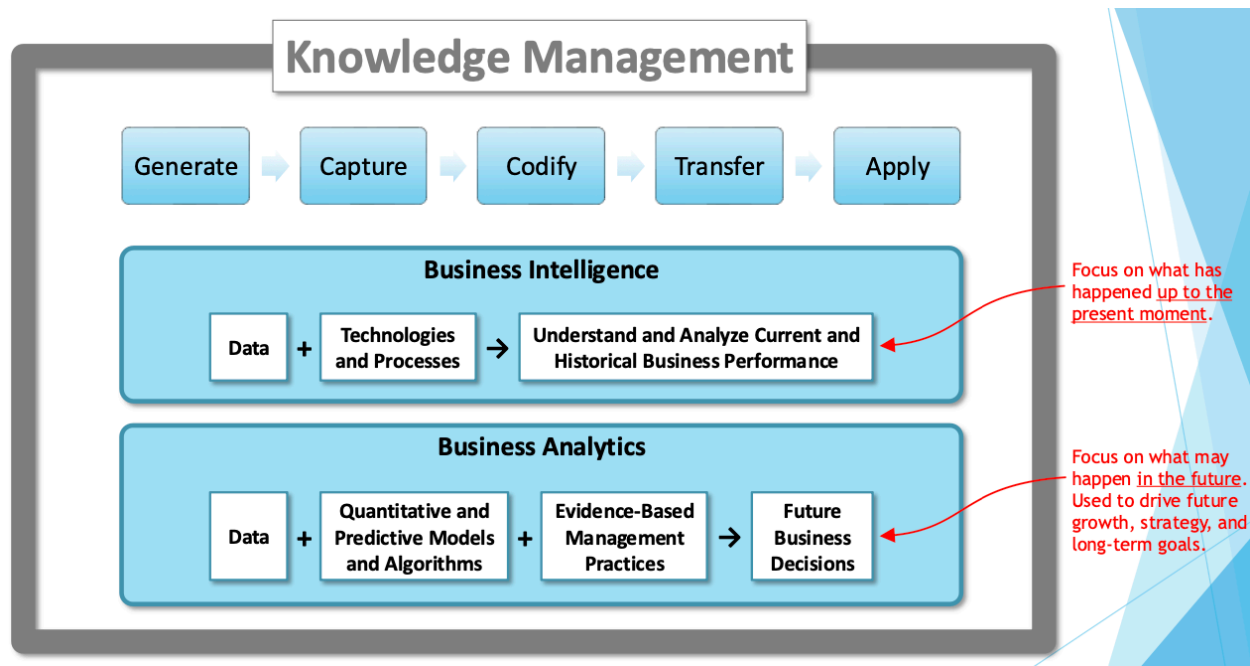
Pulling the plug

- Often projects in trouble persist long after they should have been abandoned – pull the plug!
 - Many projects are "almost done" for 50% of the project
- People can go to great lengths to sustain a doomed project when
 - A lot of time, effort, and money has been invested to date
 - High penalties for failure
 - Reputation, competitive advantage, firings
 - Gone too far to turn back
 - Continued funding depends on project
 - Emotional attachment to the project by powerful individuals

Chapter 12: Business Intelligence, AI, Knowledge Management, & Analytics

Business value with data, AI, & analytics

- Data and knowledge drive business decisions and future strategies.
- Companies seek to extract value from massive datasets on customers, competitors, and processes.
- Algorithms and analytics provides actionable insights.
- AI enables businesses to generate ideas and automate decision-making.
- The explosion of data, fueled by IoT and digital assets, requires advanced business intelligence.
- Understanding knowledge management, AI, and analytics is crucial for digital business success.



Knowledge management processes

- Knowledge must be communicated and shared to be useful.
- A company's value depends significantly on creating and managing knowledge.
- Five main processes of knowledge management:
 1. Knowledge Generation– Discovering new knowledge.
 2. Knowledge Capture– Identifying, extracting, and documenting knowledge.

3. Knowledge Codification– Structuring knowledge for easy access and transfer.
 4. Knowledge Transfer– Sharing knowledge with others for effective use.
 5. Knowledge Application– Integrating knowledge into business processes.
- Without absorption, knowledge transfer does not occur.
 - Modern AI and search tools have transformed knowledge management.

Business intelligences vs business analytics

- Business Intelligence (BI) focuses on descriptive analytics.
 - Summarizes historical and present data to show what has happened or what is currently happening.
 - Uses structured data from data warehouses and increasingly incorporates real-time data.
 - Helps managers continue successful strategies and modify ineffective ones.
 - Tools have evolved from traditional static reporting to advanced, AI-enhanced insights.
- Business Analytics (BA) focuses on predictive analytics.
 - Determines the likelihood of future outcomes using data mining, machine learning, and predictive models.
 - Incorporates big data, unstructured/semi-structured data, and data lakes.
 - Guides business planning and growth strategies by predicting trends and market shifts

Why business analytics is important

- Business analytics helps companies gain a competitive edge.
- Organizations use analytics to transform underperformance into industry leadership.
- Data-driven decision-making leads to measurable success.
- Companies competing in similar markets differentiate themselves through analytics-driven business processes.
- Analytics fuels customer personalization and operational efficiency.
- Firms that leverage analytics can maximize revenue and profitability.

How BA helps organizations

- Making information more transparent and usable at a frequency that outpaces the competition.
- Exposing variability and boosting performance by collecting and analyzing more transactional and performance data.
- More precisely tailoring products and services using better-designed segmentation and large data samples.
- Improving decision-making through experiments, forecasting, feedback, and just-in-time analysis.
- Developing the next generation of products and services more quickly using sensor data to collect after-sales information on product usage, performance, and other valuable issues.

Components of business analytics

- Data Sources - The foundation of analytics, including structured and unstructured data from internal and external sources
- Software Tools - Applications for statistical analysis, forecasting, predictive modeling, and optimization.
- Data-Driven Environment - An organizational culture that integrates analytics into decision-making and strategy.
- Skilled Workforce - Experts such as data scientists, chief data officers, and analysts who manage and interpret analytics.

Social media analytics

- The process of collecting, analyzing, and interpreting data from social media platforms.
- Uses text, images, videos, and user interactions to extract insights.
- Helps businesses understand customer behavior, brand reputation, and marketing effectiveness.
- Why Important?
 - o Understand Audience Behavior
 - o Monitor Brand Reputation
 - o Measure Campaign Effectiveness
 - o Identify Trends & Patterns
 - o Guide Business Decisions

Intellectual capital and intellectual property

- What is Intellectual Capital?
 - Valuable knowledge that is captured and used within a company.
 - An asset that gives businesses a competitive edge.
 - Includes processes, secret formulas, expertise, unique designs, and data collections.
 - Often considered the property of the organization.
 - When knowledge is explicitly documented, it becomes Intellectual Property (IP).
- How is Intellectual Property Different from Physical Property?
 1. It is nonexclusive
 2. It has a low reproduction cost

Types of IP protections in Canada

1. Copyright– Automatically granted when you create a work (e.g., books, music, software); lasts 70 years after the creator's death.
2. Trademark– Protects a brand's name, logo, slogan, or packaging; registration is optional but makes enforcement easier; lasts 10 years and is renewable.
3. Patent– Protects new inventions; must apply, and the first to file gets the patent; lasts 20 years.
4. Industrial Design– Protects the look and shape of a product (not function); must be registered and lasts 15 years.

The dark side of big data - bias & ethical concerns

1. False Discoveries and misleading correlations
2. Hidden casual factors leading to incorrect conclusions
3. AI can be unfair or discriminatory
4. Privacy concerns and consumer backlash

How to reduce Bias in AI & Big Data?

1. Recognize Bias in data and algorithms
2. Increase transparency & explainability
3. Allow users to provide feedback
4. Regularly test and adjust AI models

Chapter 13: IDK the title

Responsible computing

- Companies often encounter ethical dilemmas as they try to use their IS to create and exploit competitive advantages
- Managers must make systematic, reasoned judgements about right and wrong and take responsibility for them.
- Unfortunately, well-accepted guidelines and laws to dictate behavior do not exist, and managers are left with little direction as they face difficult decisions

Stockholder theory

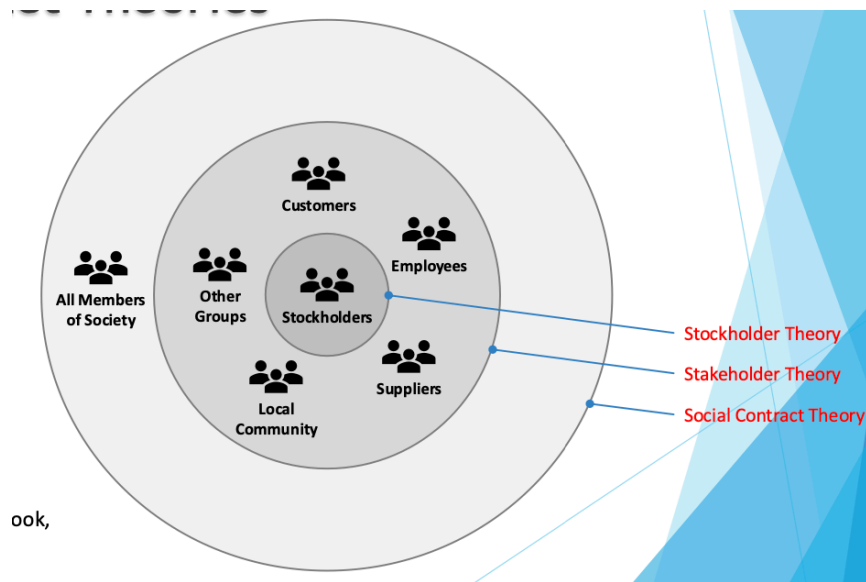
- Stockholders provide funding for a firm, expect managers to act as agents in furthering stockholders' goals.
- Managers are supposed to act in the interest of shareholders.
- Short-term sacrifices are okay if they lead to long-term gains.
- Stockholder theory qualifies the manager's duty in two salient ways
 - o Managers are bound to employ legal, nonfraudulent means
 - o Managers must take the long-term view of shareholder interest
 - They are obliged to forgo short-term gains if doing so will maximize value over the long term

Stakeholder theory

- Businesses have responsibilities not just to stockholders but to all stakeholders.
- Stakeholders include customers, employees, suppliers, community, environment.
- Managers must balance competing interests and avoid harming any one group.
- Stockholders → Want higher returns.
- Customers → Expect quality products and fair prices.
- Employees → Want fair wages, safe conditions, and job security.
- Suppliers → Seek long-term business relationship.
- Local Community → Cares about ethical business practices and environmental impact.
- When stakeholders feel that their interests haven't been considered adequately by the managers making the decisions, their only recourse may be to stop participating in the firm.

Social contract theory

- Social contract theory places responsibility on corporate managers to consider the needs of the society in which the corporation is embedded
- They should create value for society, not just profits.
- The social contract comprises two distinct components
 - o Social welfare – Benefits must outweigh costs.
 - o Justice – corporations must pursue profits legally, without fraud or deception, and avoid actions that harm society.



Responsible use of information

- Companies use big data and advanced analytics to study customer behavior.
- Businesses can now profile customers, track purchase history, and predict needs.
- These insights help companies increase sales and customer engagement.
- There are risks
 - o Target analyzed a teen girl's shopping habits and detected possible pregnancy-related purchases.

How companies use our data

- Businesses can estimate our income based on purchase habits.
- Retailers track and analyze your past purchases to suggest new products.
- Companies tempt customers with personalized discounts and limited-time offers.
- This data can help those companies who know how to use it be more competitive

Importance of data protection

- Firms have the responsibility to protect data by putting safeguards in place
 - o To protect against cyberattacks (as described in chapter 7)
 - o To prevent unauthorized access to data
 - o To use data responsibly

Data borders

- Many countries have privacy laws to protect their citizens' sensitive data.
- Companies operating globally must follow different regulations, making business more complex.
- This also serves to feed a foundation of conflict between some nations.

The four pillars of information ethics

- Developed by Richard Mason, PAPA addresses ethical concerns in Information Management.
- Why Important?
 - o Increase data collection -> higher risks for individuals and businesses.
 - o Ethical concerns as data-driven decisions impact lives.
 - o Digital economy relies on big data and AI.
- Mason suggests a number of questions that can be asked by managers (or actually any individual) to gain control over information.

Area	Critical Questions
Privacy	What data and information must people reveal about themselves to others? Can the data and information that people provide be used for purposes other than those for which the people were told that it would be used? Do people know what data and information is collected about them? Can the data people provide in multiple datasets be aggregated and processed so that they can be identified?
Accuracy	Who is responsible for the reliability, authenticity, and accuracy of data and information? Is there a plan in place to identify errors or biases? What level of inaccuracy is tolerable in a dataset? Who has the right to access IoT sensor data?
Property	Who owns the data, information, and aggregated data sets? Who owns the channels of distribution, and how should they be regulated? What is the fair price of data and information that is exchanged?
Accessibility	What data and information does a person or organization have a right to obtain, with what protection, and under what conditions? Does the person accessing personal data and information "need to know" the information that is being accessed? Will a person be harmed by choosing not to provide data or information to a service or organization?

PAPA - privacy

- The privacy paradox refers to the phenomenon where people express concern over their privacy online, yet engage in behaviors that compromise their privacy
- What contributes to this?
 - o People often prioritize convenience over privacy
 - o People are unaware of the implications of sharing their information online and how it can be used
 - o Trust that the organization they are sharing it with will safeguard it and use it responsibly
 - o Social norms and peer pressure can influence behaviour

PAPA - accuracy

- Individuals must have confidence that the information stored about them in corporate information systems is accurate
 - o Inaccurate information can lead to financial harm
 - o Or serious physical consequences if health information at a hospital is entered incorrectly or mixed up with another patient's data
- Managers can and must design controls to prevent such situations
 - o Data entry errors must be controlled and managed; data must be kept up to date
 - o Especially when it involves the potential for serious harm
 - o Hospitals have learned their lessons and now have rigorous protocols to always double-check a patient's wristband as part of every interaction
 - o The wristband also provides electronic identification and access to the person's record (for inquiry and update)
- One method of improving accuracy is to allow the users to see their own records and then to allow them to request changes if inaccuracies are found
 - o Credit reporting agencies such as TransUnion enable you to view the information they have on file and request changes if you find errors
- Applying data retention rules can also help
 - o This ensures that data is kept no longer than necessary
 - o Tagging data or data sets with "expiration" dates or using business rules to determine when data is no longer viable or relevant
 - o In some organizations the data must be destroyed or at least archived and moved offline; sometimes not easy

Manager's role in ethical information control

- Data breaches and unethical handling of information can harm customers and damage a company's reputation.
- Three Best Practices for Ethical Information Control
 - o Create a Culture of Responsibility
 - o Implement Governance Processes
 - o Avoid Decoupling

Impact of AI on business and society

- Schools at all levels
 - o Schools needed to adapt to check for AI-generated homework submissions
- Jobs
 - o Many industries have benefited from AI's efficiency, but concerns remain about job losses.
 - o Intelligent automation has the potential to replace humans.
- Schools at all levels
 - o Schools needed to adapt to check for AI-generated homework submissions
- Jobs
 - o Many industries have benefited from AI's efficiency, but concerns remain about job losses.
 - o Intelligent automation has the potential to replace humans.
 - o Employers have a moral responsibility to their workers whose jobs are affected by AI to provide alternative jobs or retraining.

Green computing and its driving factors

- Green computing is the use of computers and other computing devices and equipment in energy-efficient and eco-friendly way
- Factors Driving Green Computing
 - Lower Energy Costs
 - Climate & Environmental Responsibility
 - Government Regulations related to Energy Conservation